

Learning from Dialogue Traces: Retrieval-Guided Tool Use and Response Generation for Task-Oriented Dialogue Agents

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Abstract

Effective task-oriented dialogue (TOD) requires agents to coordinate two tightly coupled decisions of when to invoke external tools and how to formulate a response that satisfies the user’s goal. However, their turn-level behavior remains weakly grounded under fixed zero-shot prompts. Meanwhile, existing retrieval-based TOD agents primarily retrieve task-relevant content from external knowledge sources, guiding which information to include but offering limited guidance on how to select tools and formulate responses in similar dialogue contexts. To mitigate this, we introduce **TRACE**, a **Turn-level Retrieval Agent with Contextual Experiences for TOD**¹. Instead of relying on retrieved factual records alone, TRACE retrieves goal-conditioned dialogue snippets as behavioral demonstrations at each turn. These snippets capture similar user goals, constraints, tool-use decisions, and response patterns to guide both tool calls and response generation. Experiments on MultiWOZ 2.1 and Schema-Guided Dialogue (SGD) show that TRACE improves Inform and Success over a competitive LLM agent and end-to-end fine-tuned models under both simulation-based and static-user evaluations. Human evaluation and retrieval-focused analyses further show that goal-conditioned snippets and retrieval quality drive these gains.

1 Introduction

Task-oriented dialogue (TOD) systems assist users in completing goal-driven tasks, such as booking hotels, reserving restaurant tables, and planning travel. Early TOD systems (Wu et al., 2020; Lee et al., 2021; Lin et al., 2021; Peng et al., 2020) typically followed a modular pipeline, with separately trained components for Natural Language Understanding (NLU), Dialogue State Tracking (DST), Policy Learning (POL), and Natural Language Generation (NLG). While modularity en-

ables component-wise specialization, it requires extensive engineering and suffers from error propagation. More recent work has shifted toward end-to-end fine-tuned models that jointly predict intermediate states and responses (Ham et al., 2020; Hosseini-Asl et al., 2020; Yang et al., 2021; Lee, 2021; Sun et al., 2022; Imrattanatrai and Fukuda, 2023). However, these models depend on template-heavy generation, which makes handling open-vocabulary values and domain transfer difficult.

Recent developments in large language models (LLMs) offer a more flexible alternative, as they can follow instructions and generate fluent responses with little to no task-specific training (Yao et al., 2023; Shen et al., 2023; Lu et al., 2023). In TOD, AutoTOD (Xu et al., 2024) introduces an autonomous zero-shot LLM-based agent that decides whether to query task databases via tools/APIs before generating a response. Meanwhile, other works on TOD agents have further incorporated explicit retrieval mechanisms to access external information, such as entity records, user intents, or task-specific knowledge (Du et al., 2026; Zhang et al., 2023; Chen et al., 2025). However, these approaches primarily retrieve task-relevant content rather than examples of how similar dialogue turns were handled. Thus, they help determine what information is needed to satisfy the user goal, but provide limited explicit guidance on how the agent should act and respond in similar contexts.

To address this limitation, we propose **TRACE**, a **Turn-level Retrieval Agent with Contextual Experiences**. Unlike prior retrieval-augmented TOD agents that primarily retrieve factual content from external databases, TRACE retrieves past dialogue-turn experiences as behavioral demonstrations. Specifically, TRACE performs *per-turn* retrieval of dialogue snippets conditioned on the current user utterance and predicted belief state. These snippets reflect similar user goals, dialogue constraints, tool-use decisions, and response pat-

¹<https://github.com/aistairc/TRACE-TOD>

terns, thereby guiding both tool selection and response generation. In this way, TRACE shifts retrieval from task-content grounding to agent-policy guidance. Our contributions are threefold. First, we introduce TRACE, which integrates belief-conditioned turn-level retrieval into a TOD agent to retrieve goal-consistent dialogue demonstrations for guiding tool calls and responses. Second, we evaluate TRACE under both simulation-based and static-user settings on two widely used benchmarks, MultiWOZ 2.1 and SGD, showing consistent improvements over a competitive agent and end-to-end fine-tuned baselines. Third, we complement automatic evaluation with human judgments and retrieval-focused analyses, identifying when retrieval helps or hurts and clarifying its role in improving agent reliability across diverse scenarios.

2 Related Works

End-to-End TOD systems Early end-to-end fine-tuned TOD models used autoregressive LMs such as GPT-2 (Radford et al., 2018) to unify dialogue component generation into a single sequence. Ham et al. (2020) pioneered this direction by sequentially modeling user utterances, belief states, system actions, and delexicalized responses; SimpleTOD (Hosseini-Asl et al., 2020) then integrated DB results to improve response relevance. Building upon these foundations, SOLOIST (Peng et al., 2021) incorporated contrastive learning, while UBAR (Yang et al., 2021) modeled full dialogue sessions as part of the input context. Beyond GPT-2, T5 (Raffel et al., 2020) has been leveraged; for example, Lee (2021) used a dual-decoder T5 to jointly predict multiple dialogue components, while Sun et al. (2022) introduced reconstruction-based objectives to enhance generation robustness. Wang et al. (2022) argued against a delexicalization process and only utilized natural language components. GALAXY (He et al., 2022) introduced a dialogue action prediction objective and uses consistency regularization to learn policies, while MARS (Sun et al., 2023) explored contrastive learning for belief/action state representations. Bang et al. (2023) proposed TOATOD, with adapters that can learn independently for each task on a pre-trained model. To handle unseen domains, SchemaTOD (Imrattanatrai and Fukuda, 2023) addressed domain generalization via schema-level context. Although these end-to-end systems simplify TOD modeling, they require large-scale,

domain-specific training data. Despite their unified modeling, they struggle to generalize to unseen domains or adapt to new dialogue scenarios.

LLM-based Agent LLMs enable agents to infer goals, call external APIs, and generate responses with minimal task-specific training. Representative frameworks such as ReAct (Yao et al., 2023), HuggingGPT (Shen et al., 2023), and Chameleon (Lu et al., 2023) demonstrate the flexibility of LLM agents in planning, reasoning, and tool usage for various tasks. In TOD, AutoTOD (Xu et al., 2024) introduced an LLM-based agent that interprets user utterances and dialogue history to select tools and compose responses. AutoTOD demonstrates strong zero-shot capabilities; however, it relies on the LLM’s internal reasoning to decide tool use and response generation. More recently, some TOD agents have incorporated retrieval-based mechanisms to access relevant information from external databases, such as entity records or task-specific knowledge (Du et al., 2026; Zhang et al., 2023; Chen et al., 2025). While these approaches use retrieval primarily as a source of factual content, our work explores a different role for retrieval in TOD agents. Inspired by prior work on retrieving past experiences to support agent decision-making (Humphreys et al., 2022; Zhao et al., 2024; Kagaya et al.; King and Flanigan, 2023), our agent retrieves semantically relevant dialogue snippets from past interactions. These snippets provide examples of response and tool-use patterns in similar dialogue contexts, thereby guiding the agent’s current tool calls and responses.

3 Methodology

Problem Definition In TOD systems, the goal is to help users complete specific tasks. Given a dialogue history $H_t = \{(u_1, r_1), \dots, (u_{t-1}, r_{t-1})\}$, where u_i and r_i denote the user utterance and system response, the agent must determine actions (*i.e.*, tool calls) a_t and generate a response r_t based on the user utterance u_t and belief state b_t at turn t .

The proposed agent, **TRACE**, is a retrieval-augmented LLM agent designed to ground tool use and response decisions. At each dialogue turn t , TRACE retrieves a set of relevant dialogue turn snippets, $\mathcal{R}_t = \{S_1, S_2, \dots, S_k\}$, and injects them into the task-specific prompt as in-context demonstrations. Each snippet S_i consists of a user utterance u_i , belief state b_i , system action a_i , and response r_i , thereby providing examples of how sim-

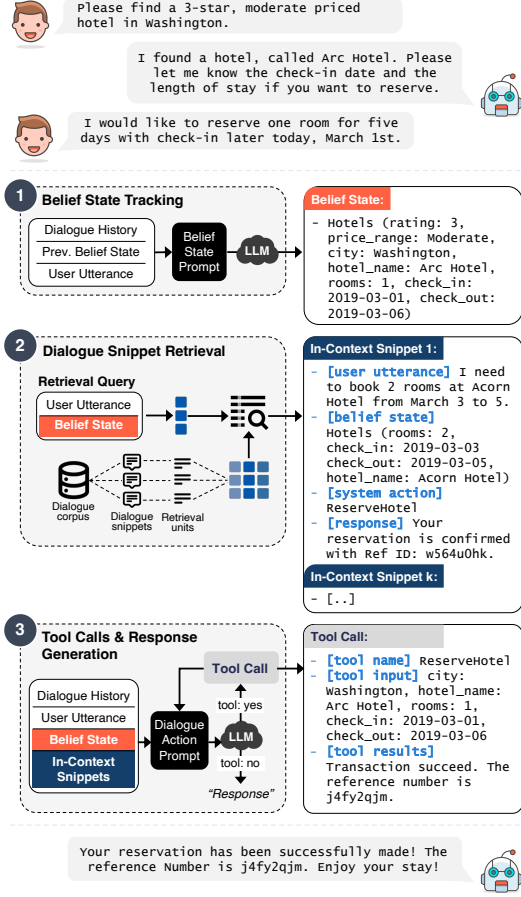


Figure 1: Three-stage process of TRACE: (1) belief state tracking to capture the user’s intent, (2) retrieval of relevant dialogue snippets using the user utterance and belief state to guide tool use and response grounding, and (3) tool calls and response generation.

ilar dialogue contexts were handled. By leveraging these retrieved snippets, TRACE guides both tool calls and response generation, allowing the agent to ground its decisions in relevant past dialogue experiences. As illustrated in Figure 1, TRACE operates in three stages: (1) the agent performs *Belief State Tracking* to track the user’s intent, (2) in *Dialogue Snippet Retrieval*, the agent forms a retrieval query using the user utterance and belief state to retrieve dialogue snippets, and (3) in *Tool Calls and Response Generation*, the agent is grounded by the retrieved snippets and dialogue context to decide whether to invoke tools (e.g., database lookup, booking) or generate the response.

Belief State Tracking Given the user utterance u_t at turn t , the agent performs belief state tracking by generating a set of domain-specific slot-value pairs that represent the user’s constraints. We adopt the common format used in the previous works, e.g., hotel (area: centre, price:

cheap, internet: yes), to maintain compatibility with the pre-trained knowledge of LLMs (Peng et al., 2021; Hosseini-Asl et al., 2020; Yang et al., 2021). In TRACE, belief state tracking is handled by prompting an LLM with the dialogue history H_t , prior belief state b_{t-1} , and user utterance u_t . The model is prompted with a structured template, which includes: (1) a task overview description, (2) step-by-step instructions, (3) an explicit output format, (4) valid schemas with available domains and their corresponding slots, (5) the input section which provides the necessary dialogue context, and (6) the output section which instructs the model to generate the belief state b_t .

While the recent LLM-based agent (Xu et al., 2024) omits belief states from the dialogue context, this design choice introduces notable limitations. Without an explicit representation of the user’s evolving preferences across domains, the agent can easily lose track of previously stated constraints or generate redundant or incorrect tool-call requests. In contrast, maintaining an explicit belief state enables TRACE to summarize and accumulate user preferences over turns. This leads to more consistent decision-making and more robust behavior. Furthermore, belief states serve as effective anchors for retrieving relevant dialogue snippets, as their structure helps the agent interpret the current dialogue context more effectively and connect it to semantically similar dialogue turns.

Dialogue Snippet Retrieval To enhance response generation, TRACE retrieves per-turn snippets from a dialogue corpus to serve as in-context demonstrations for the dialogue action prompt.

To retrieve the dialogue snippets, we perform dense retrieval using FAISS (Johnson et al., 2019), a fast vector similarity search library. Given a dialogue corpus consisting of n past snippets (i.e., every dialogue turn in a corpus), $C = \{S_1, S_2, \dots, S_n\}$, we define each retrieval unit U_i as a concatenation of u_i and b_i . These retrieval units are embedded into a shared semantic space using a pre-trained embedding model. These result in a set of vectors, $\{\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n\}$. At runtime, the retrieval query q_t is constructed by concatenating the user utterance u_t with the predicted belief state b_t , and encoded into a vector $\vec{q}_t$ in the same embedding space. We then compute the cosine similarity between $\vec{q}_t$ and each $\vec{u}_i$. The top- k most similar retrieval units are selected. Based on each retrieved unit, we retrieve the full dialogue turn

snippet $S_i=(u_i, b_i, a_i, r_i)$, which is then injected into the dialogue action prompt at the next stage.

This retrieval-augmented prompting enables the agent to access semantically aligned dialogue turns and observe how similar dialogue turns were handled, thereby improving the agent’s decision-making and response quality. Crucially, the dynamic retrieval mechanism supports generalization to diverse dialogue scenarios.

Tool Calls and Response Generation In the final stage, the agent performs tool calls and generates responses based on the current dialogue context and retrieved in-context snippets via the dialogue action prompt. We adopt the *ReAct* paradigm (Yao et al., 2023), which enables the agent to reason about the user’s request before taking action. We implement the task using a format similar to that of the existing work, AutoTOD (Xu et al., 2024), in which the LLM agent is guided by a *scenario description* followed by *task information* to decide the next action. However, unlike AutoTOD, TRACE augments this prompt with in-context demonstrations retrieved from past dialogues. These retrieved snippets are injected into the prompt to illustrate how similar scenarios were previously handled. This retrieval-augmented prompting provides grounded examples that help the agent make more informed decisions. Moreover, to guide the agent appropriately, we modified the scenario description to explicitly instruct it to treat the retrieved snippets as references, rather than as sources of factual content to be reused directly.

Given the constructed dialogue action prompt, the agent first decides whether to invoke a tool or generate a response. If a tool call is required, the agent must submit the appropriate tool request by specifying the tool name and structured input arguments. Once the tool result is obtained, it is appended to the dialogue context. This reasoning loop continues until the agent determines that no further tool calls are needed, at which point it generates a system response grounded in the full dialogue history, the updated belief state, retrieved dialogue snippets, and the tool outputs.

4 Experiments

Datasets We use two benchmark datasets for multi-domain TOD systems: MultiWOZ 2.1 (Budzianowski et al., 2018), hereafter MultiWOZ, and the Schema-Guided Dialogue (SGD) dataset (Rastogi et al., 2020), which includes dia-

logues across multiple services and domains. To keep the experimental setup lightweight, we limit the number of SGD test samples to 1,000. We report other statistics in the Appendix.

Evaluation Metrics We adopt standard metrics for TOD: **Inform**: Measures whether the system provides correct entities that match the user’s constraints; **Success**: Measures whether the system successfully provides all requested attributes; and **Book**: For the MultiWOZ dataset, this metric evaluates whether the system completes a reservation by issuing a booking with a valid reference number.

To evaluate these metrics in a more interpretable, context-aware manner as proposed by (Xu et al., 2024), we use an LLM-based evaluator, specifically GPT-3.5. For **Inform** (I), the evaluator extracts the entity mentioned in the system response, queries the database for the full set of attributes, and checks whether they match the ground truth user’s goal constraints. For **Success** (S), it verifies that all required information has been accurately provided to the user. For **Book** (B), the evaluator checks whether a booking reference is issued and whether it satisfies the ground truth user’s booking constraints. Finally, we report the **Combined** (C) score as a weighted aggregation of the three metrics as $0.5 \times \text{Inform} + 0.25 \times (\text{Success} + \text{Book})$, to reflect the task success, goal satisfaction, and booking execution. For scores achieved by TRACE, we performed a statistical analysis using a paired t-test with Benjamini-Hochberg correction. Statistical significance is determined at $p < 0.05$. Additionally, we conduct human evaluation on a subset of dialogues from the main experiments.

Experimental Settings We employ open-sourced models: Llama3.1-70B-Instruct and Qwen2.5-72B-Instruct, as well as proprietary models from OpenAI²: gpt-3.5-turbo-0125, and gpt-4o-mini-2024-07-18, as backbone models. For the retrieval strategy, we set k , the number of relevant dialogue snippets, to 3 and utilize OpenAI’s text-embedding-ada-002 as the default embedding model. However, we conducted ablation studies on the effect of k and different embedding models. For the dialogue corpus, we prepared it using each dataset’s training dialogues. In the case of MultiWOZ, where ground-truth system actions are unavailable, each dialogue snippet is constructed without the true system

²<https://openai.com/blog/openai-api>

Method	Domain Level				Dialogue Level			
	I	S	B	C	I	S	B	C
SimpleTOD*	32.5	29.4	-	23.6	18.8	22.0	-	14.9
UBAR*	40.8	33.3	-	28.7	24.0	26.8	-	18.7
GALAXY*	44.4	35.1	-	31.0	26.4	28.8	-	20.4
Mars*	42.7	34.4	-	30.0	25.9	27.5	-	19.8
Mars (5%-shot)*	28.9	26.3	-	21.0	16.2	14.0	-	11.6
TOATOD*	45.3	36.7	-	31.8	27.8	26.9	-	20.6
AutoTOD								
Llama-3.1-70B	60.0	50.3	53.5	55.8	44.6	41.0	51.9	44.6
Qwen2.5-72B	81.0	60.3	84.5	75.5	68.0	50.1	82.0	65.5
	83.0	68.1	91.5	81.4	83.7	69.5	93.1	82.5
GPT-3.5	62.6	39.4	56.9	54.5	45.9	30.5	53.9	42.5
GPT-4o-mini	67.7	48.7	67.7	61.9	49.7	37.3	64.6	48.6
	69.1	42.6	72.3	63.3	68.3	49.5	69.4	62.8
TRACE								
Llama-3.1-70B	62.7 [†]	55.5 [†]	55.3	59.1 [†]	46.9	47.9 [†]	52.2	48.1 [†]
Qwen2.5-72B	82.4	62.7	84.9	76.9	68.8	51.0	82.7	66.3
	89.4	70.2	95.7	86.2	89.7	71.0	97.2	85.8
GPT-3.5	65.7 [†]	50.6 [†]	59.8	60.0 [†]	48.5	41.3 [†]	56.3	47.4 [†]
GPT-4o-mini	72.9 [†]	56.3 [†]	71.4 [†]	67.6 [†]	56.5 [†]	45.9 [†]	68.3	55.0 [†]
	76.6	63.8	85.1	75.5	76.0	67.1	81.9	75.0

Table 1: Results on MultiWOZ. * indicates results reported in prior work (Xu et al., 2024). † denotes significantly better scores than those achieved by AutoTOD, with the same LLM. Blue-highlighted scores are based on human evaluation of 50 dialogues.

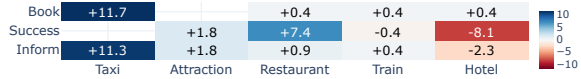


Figure 2: Gains per domain on MultiWOZ by TRACE over AutoTOD using Qwen2.5.

actions. Experiments with open-source models are conducted on an H200 GPU with 141GB of memory. All reported results are from a single run.

We consider two evaluation settings. First, **Interactive Simulation (LLM-as-User)**, following the AutoTOD framework, simulates the user side based on dialogue goals and the current dialogue history using an LLM, except for the first dialogue turn, to better reflect real-world scenarios. Second, **Static Evaluation (User-Given)**, in this case, the system is provided with the ground-truth user utterance at every dialogue turn. This setting ensures a fair comparison across models; however, it does not enforce dialogue consistency when the system deviates from the expected path. Thus, it is more difficult to assess whether the systems truly performed well without redirecting them using ground-truth user utterances. In the simulation setting, we use the same LLM for the agent and the user simulator to maintain a consistent interaction.

4.1 Experimental Results

Results on MultiWOZ We evaluate our TRACE against the existing end-to-end fine-tuned systems

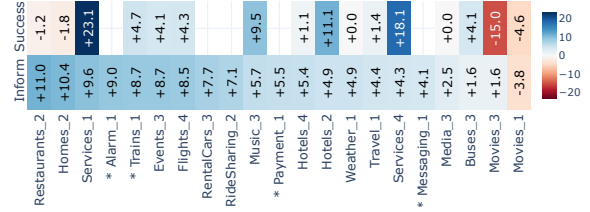


Figure 3: Gains per service on SGD by TRACE over AutoTOD using Qwen2.5. * indicates unseen services where past dialogues are unavailable.

and LLM-based agent on the MultiWOZ benchmark (Table 1). At both domain and dialogue levels, TRACE consistently outperforms AutoTOD using the same LLM, as well as the end-to-end fine-tuned approaches. Notably, TRACE with Qwen2.5 achieves the highest scores on all metrics. When using Llama-3.1, GPT-3.5, and GPT-4o-mini, TRACE achieves significant improvements in Inform, Success, and Combined scores relative to AutoTOD at the domain level. At the dialogue level, the three LLMs also exhibit strong performance with significant improvements in Success and Combined scores. Similarly, in human assessments on two strong backbones (*i.e.*, Qwen2.5 and GPT-4o-mini), TRACE consistently outperforms AutoTOD. These results suggest that incorporating retrieval-augmented dialogue snippets greatly enhances the agent’s grounding for task completion.

Figure 2 further illustrates the performance gain of TRACE over AutoTOD when using the Qwen2.5 model across five domains in MultiWOZ. TRACE achieves performance gains across most domains, particularly in the Taxi domain, where it significantly outperforms AutoTOD in both Inform and Book scores, and in the Restaurant domain, where it achieves a high Success gain. However, TRACE also achieves performance losses in the Hotel domain. Further analysis reveals that the Hotel domain involves a more heterogeneous slot set than other domains, *e.g.*, boolean attributes such as parking/internet, categorical attributes like type/stars, and booking fields like stay length. Therefore, errors in belief state tracking or retrieval-induced slot conflicts are more likely to cause the selected entity to be flipped and penalized under strict Inform and Success metrics, which require that all requested slots must be satisfied.

Results on SGD We evaluate TRACE on the SGD dataset to assess its capability in a multi-service, multi-domain setting. Table 2 reports re-

Method	Intent Level		Service Level		Dialogue Level	
	I	S	I	S	I	S
SimpleTOD*	–	–	13.1	10.4	9.3	9.5
ZS-TOD*	–	–	24.9	11.2	18.3	10.1
AutoTOD						
Llama-3.1-70B	69.5	27.7	66.6	25.2	52.3	22.2
Qwen2.5-72B	67.7	21.2	64.6	18.5	48.6	15.5
	–	–	70.8	51.7	56.0	50.0
GPT-3.5	40.8	15.4	38.7	13.2	24.9	12.3
GPT-4o-mini	61.1	19.5	59.0	16.9	42.5	13.6
	–	–	66.0	50.0	46.0	52.4
TRACE						
Llama-3.1-70B	66.7	26.3	62.7	23.5	47.6	18.5
Qwen2.5-72B	73.5 [†]	25.5 [†]	70.4 [†]	22.7 [†]	54.2 [†]	17.6
	–	–	71.0	56.9	52.0	54.8
GPT-3.5	45.5 [†]	15.7	42.2 [†]	13.0	27.2	12.3
GPT-4o-mini	73.0 [†]	21.7 [†]	71.0 [†]	18.2	55.3 [†]	14.6
	–	–	70.8	51.7	54.0	52.4

Table 2: Results on SGD. * indicates results reported by prior work (Xu et al., 2024). † indicates statistically significant improvement over AutoTOD (with the same LLM). Blue-highlighted scores come from human evaluation on 50 dialogues.

sults across three levels. TRACE with all LLMs, except for Llama-3.1, outperforms AutoTOD when using the same LLM. For instance, with Qwen2.5, TRACE achieves the highest Inform at the intent level, along with improved service- and dialogue-level Inform. The improvements of Qwen2.5 are statistically significant relative to AutoTOD, except for Success at the dialogue level. With GPT-4o-mini, TRACE improves upon AutoTOD by large margins, particularly in service- and dialogue-level Inform scores. Also, with GPT-3.5, TRACE achieves better performance across nearly all metrics. With human evaluation, TRACE consistently outperforms AutoTOD when both use the same underlying LLM across most metrics. However, the gains are noticeably smaller than on MultiWOZ, and for one setting (*i.e.*, dialogue-level Inform with Qwen2.5), TRACE underperforms AutoTOD. This pattern highlights the greater difficulty of the multi-service setting, where users’ goals span multiple schemas, and the agent must maintain a consistent state across services.

Figure 3 highlights the benefits of dialogue snippet retrieval for individual services. TRACE achieves substantial gains over AutoTOD’s performance in Inform across nearly all services. Improvements are particularly strong in services such as Restaurants_2 and Homes_2, where gains in Inform rates exceed 10 absolute points. In terms of Success rates, TRACE also shows improvements in many services. Strong gains are observed for Services_1, Services_4, and Hotel_2. Interestingly,

Method	Domain Level				Dialogue Level			
	I	S	B	C	I	S	B	C
AutoTOD								
Llama-3.1-70B	46.8	37.9	43.0	43.3	27.7	29.5	38.7	30.0
Qwen2.5-72B	74.7	53.7	76.3	68.7	58.0	43.3	73.1	56.1
GPT-3.5	55.1	31.3	43.8	45.7	34.2	23.0	40.4	31.1
GPT-4o-mini	58.7	41.2	48.1	51.3	36.5	31.8	43.8	35.5
TRACE								
Llama-3.1-70B	51.9 [†]	41.8 [†]	43.6	47.2 [†]	32.8 [†]	32.3	40.1	33.7 [†]
Qwen2.5-72B	77.1 [†]	56.2	79.1 [†]	71.2 [†]	61.7 [†]	46.0	76.2 [†]	59.5 [†]
GPT-3.5	59.2 [†]	40.4 [†]	49.0 [†]	51.5 [†]	38.7 [†]	30.2 [†]	45.2 [†]	36.6 [†]
GPT-4o-mini	62.8 [†]	44.6 [†]	59.4 [†]	56.6 [†]	41.8 [†]	35.7 [†]	55.1	42.0 [†]

Table 3: Results using ground-truth user utterances on MultiWOZ. † denotes statistically significantly better scores than those by AutoTOD, under the same LLM.

the system also improves performance on unseen services (*e.g.*, Alarm_1, Messaging_1, Payment_1, Trains_1), suggesting that TRACE can effectively generalize by leveraging contextually similar dialogues. Although a few cases (*e.g.*, , Movies_1 and Movies_3) exhibit drops in Inform and/or Success, we observe that these are largely caused by long-tail, open-vocabulary values in Movies-specific slots (*e.g.*, movie, theater, cast, and director names). In such cases, both the model and the retrieval component are more likely to propagate incorrect strings into the query or the generated response.

Results with Static User Utterances To further validate the effectiveness of TRACE, we conducted the evaluation using static ground-truth user utterances from the MultiWOZ test set, as shown in Table 3. Across all LLMs, TRACE consistently improves upon AutoTOD. When paired with Qwen2.5, TRACE achieves the highest scores across all evaluation metrics, demonstrating significant gains over AutoTOD in Inform, Book, and Combined scores. Similarly, with Llama-3.1, GPT-3.5, and GPT-4o-mini, TRACE also leads to statistically significant improvements in almost all metrics. Compared with results obtained with simulated user utterances, overall scores in the static setting are consistently lower. This is expected because the static setting prevents the user from adapting to system behavior. Nonetheless, TRACE continues to deliver substantial improvements over the baseline, which demonstrates the robustness of retrieval-augmented prompting for the TOD agent.

4.2 Ablation Studies and Analysis

We conducted ablation studies on 100 randomly sampled dialogues from MultiWOZ to address the following key research questions.

Method	Domain Level				Dialogue Level			
	I	S	B	C	I	S	B	C
Qwen2.5								
AutoTOD	82.7	58.4	86.3	78.0	70.0	52.9	82.9	69.0
AutoTOD-belief	80.6	52.8	82.4	74.6	66.0	44.3	77.6	64.2
TRACE-fixed	82.2	50.6	85.3	75.7	69.0	42.9	82.9	66.2
TRACE-random	81.7	49.4	79.4	73.6	70.0	42.9	77.6	65.2
TRACE-ME5	83.8	61.8	87.3	79.6	70.0	54.3	82.9	69.8
TRACE-ADA	83.2	64.0	85.3	79.3	70.0	57.1	81.6	70.2
TRACE-Qwen3	81.2	59.6	82.4	76.4	68.0	52.9	78.9	67.2
GPT-4o-mini								
AutoTOD	66.5	43.8	68.6	61.8	48.0	37.1	61.8	48.0
AutoTOD-belief	70.2	51.7	66.7	64.9	51.0	47.1	57.9	51.5
TRACE-fixed	71.2	50.6	73.5	67.0	51.0	41.4	68.4	53.5
TRACE-random	73.3	50.6	72.5	67.8	53.0	40.0	67.1	52.5
TRACE-ME5	66.0	48.3	65.7	61.8	50.0	41.4	60.5	50.2
TRACE-ADA	72.8	55.1	71.6	68.3	58.0	50.0	64.5	57.8
TRACE-Qwen3	75.4	61.8	73.5	71.7	58.0	55.7	67.1	59.0

Table 4: Results using different embedding approaches: text-embedding-ada-002 (ADA), multilingual-e5-large-instruct (ME5), and Qwen3-Embedding-0.6B (Qwen3), and using fixed or random snippets for TRACE.

Do Per-Turn Retrieval of Snippets Matter?

Fixed vs. Random vs. Retrieved Table 4 analyzes how TRACE benefits from different snippet strategies by comparing: (i) no snippets (AutoTOD), (ii) AutoTOD with explicit belief-state tracking (AutoTOD-belief), and (iii) TRACE with snippets selected by *random*, *fixed*, or *retrieval*-based strategies using different embedding models. From this table, we discover two important insights. First, **snippets are not universally helpful**. With Qwen2.5, injecting non-targeted snippets (fixed or random) often fails to improve over the original AutoTOD, suggesting that irrelevant demonstrations can distract the agent and destabilize action patterns. However, GPT-4o-mini is more tolerant: even noisy snippets can improve some metrics, indicating greater robustness to irrelevant context. Second, across both backbones, **retrieval-based selection is the most reliable strategy**, consistently outperforming fixed and random variants. This supports the intended behavior of TRACE as an effective dynamic few-shot agent, as its performance improves when in-context demonstrations are semantically relevant. Moreover, belief state tracking alone (AutoTOD-belief) shows mixed effects: it is beneficial for GPT-4o-mini but not for Qwen2.5, whereas belief-conditioned retrieval remains consistently beneficial. Finally, the embedding model choice affects downstream performance, as different encoders capture semantics differently, thereby altering how well retrieved snippets match the intent and slot constraints. As a result, ME5 and ADA yield the strongest overall trade-offs for Qwen2.5

Method		Domain Level				Dialogue Level			
		I	S	B	C	I	S	B	C
Qwen2.5	TRACE	83.2	64.0	85.3	79.3	70.0	57.1	81.6	70.2
	$\Delta \neg U$	-0.5	-7.8	0.0	-2.1	-2.0	-7.1	0.0	-2.2
	$\Delta \neg B$	-2.0	-7.8	-4.9	-4.2	-2.0	-7.1	-4.0	-3.7
GPT-4o-mini	TRACE	72.8	55.1	71.6	68.3	58.0	50.0	64.5	57.8
	$\Delta \neg U$	-5.8	-4.5	-6.9	-5.7	-9.0	-5.7	-7.9	-7.6
	$\Delta \neg B$	-13.1	-11.3	-26.5	-16.2	-15.0	-14.3	-27.7	-18.8

Table 5: Ablation study on retrieval query components for TRACE. We compare performance when removing the user utterance ($\neg U$) or belief state ($\neg B$) from the retrieval query. Δ indicates the change in performance compared to full TRACE.

(ADA favors Success, while ME5 favors Inform and Book), whereas Qwen3 embeddings work best for GPT-4o-mini, likely due to better alignment between the embedding space and the backbone’s interpretation of demonstrations, leading to more consistently relevant in-context examples.

Which Retrieval Query Components Matter More: Utterance vs. Belief?

Having established that dynamic retrieval of snippets is important, we next investigate which information is most useful for retrieving them. We therefore ablate the retrieval query in TRACE by removing either the user utterance ($\neg U$) or the belief state ($\neg B$), and report the changes from full TRACE in Table 5. Removing the belief state consistently degrades performance across all metrics, especially with GPT-4o-mini. Meanwhile, eliminating the user utterance from the query also degrades performance, but to a lesser extent, primarily on Success and Combined. These results suggest that belief state conditioning is a primary driver of snippet retrieval, whereas the utterance provides complementary semantic cues for matching interaction patterns within the appropriate task context. Overall, this ablation study validates our design of the TRACE retrieval module, where both user utterances and belief states jointly contribute to retrieving effective dialogue snippets. The belief state, in particular, proves essential for maintaining task grounding across dialogue turns.

How Belief State Accuracy Affects Retrieval and End Performance?

To examine how belief state accuracy impacts snippet retrieval, we compare retrieval conditioned on model-predicted versus oracle (true) belief states (Table 6). The results show an essential finding: true belief states are not a consistent upper bound for TRACE. For Qwen2.5, predicted beliefs improve domain-level

Method	Belief	Domain Level				Dialogue Level			
		I	S	B	C	I	S	B	C
Qwen2.5	True	74.9	52.8	72.5	69.1	61.0	44.3	65.8	58.5
	Pred.	75.9	56.2	72.5	70.4	58.0	50.0	65.8	57.0
GPT-4o-mini	True	64.9	43.8	63.7	59.7	46.0	37.1	52.6	45.0
	Pred.	63.4	43.8	57.8	57.3	46.0	40.0	50.0	45.8

Table 6: Performance comparison using static ground-truth user utterance between TRACE that uses either true or predicted belief states for retrieving snippets.

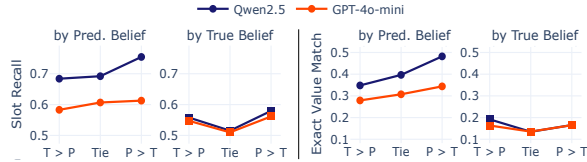


Figure 4: Retrieval-alignment metrics based on TRACE’s performance when using either predicted or true belief states by Qwen2.5 and GPT-4o-mini. X-axis denotes three different outcome cases defined by whether the dialogue-level Combined score was higher with true belief states ($T > P$), tied (Tie), or higher with predicted belief states ($P > T$).

scores, but dialogue-level outcomes diverge, as true beliefs yield higher Inform and Combined scores, whereas predicted beliefs yield higher Success. For GPT-4o-mini, true beliefs improve domain-level scores, while predicted beliefs are slightly better on dialogue-level Success and Combined scores. To explain why more accurate beliefs do not always improve downstream performance, we analyze the retrieved dialogue snippets themselves. Figure 4 measures retrieval alignment using two diagnostics that directly reflect whether retrieved demonstrations are usable for the current turn: 1) slot recall@ k that measures the fraction of query’s domain-slot constraints covered by the union of the those in top- k snippets, and 2) exact value match@ k which presents the fraction of query’s domain-slot-value constraints whose values are exactly matched in that same top- k union. Across both backbones, and most clearly for Qwen2.5, retrieval by predicted beliefs achieves higher slot recall and exact value match, particularly in the $P > T$ subset, where predicted beliefs outperform true beliefs on dialogue-level Combined score. This suggests that predicted beliefs can serve as a more effective retrieval cue by retrieving demonstrations with higher value-level compatibility, even when the belief state itself is imperfect. A plausible explanation for this is that true belief states are often more specific with more constraints. Thus, it is

Model	k	Domain Level				Dialogue Level			
		I	S	B	C	I	S	B	C
Qwen2.5	1	81.7	57.3	81.4	75.9	68.0	52.9	77.6	67.2
	3	83.2	64.0	85.3	79.3	70.0	57.1	81.6	70.2
	5	80.1	52.8	79.4	73.6	65.0	45.7	75.0	63.0
	10	79.6	58.4	80.4	74.9	62.0	48.6	75.0	62.5
GPT-4o-mini	1	69.1	51.7	67.6	64.7	52.0	44.3	59.2	52.2
	3	72.8	55.1	71.6	68.3	58.0	50.0	64.5	57.8
	5	75.9	62.9	72.5	72.0	60.0	57.1	68.4	61.0
	10	70.2	55.1	64.7	65.2	53.0	45.7	56.6	51.5

Table 7: Results on different top- k retrieved snippets.

harder for the retriever to find many dialogue snippets that satisfy the same combination of slots and values. As a result, the top- k retrieved set covers only parts of the relevant slots and values, making the demonstrations less directly applicable. Overall, these results indicate that retrieval quality depends not only on belief state accuracy but also on whether snippets contain salient slots and values to guide reliable response generation. This motivates a future direction for improving snippet retrieval for downstream TOD success.

How Shorter and Longer Retrieved Snippets Affect the Performance?

We examine the impact of varying the number of retrieved snippets (k). Table 7 shows that retrieving too few or many dialogue snippets does not improve TRACE and that the best k is different depending on the model. With Qwen2.5, the best results are obtained with a $k=3$, while decreasing k to 1 and increasing k to 5 or 10 degrade all metrics. In contrast, GPT-4o-mini benefits from an increase to $k=5$ on both Domain and Dialogue levels, but performance drops sharply at $k=10$. Thus, we infer that too few snippets may under-support models, whereas too many snippets increase prompt length and noise, which prevent the model to produce consistent actions.

5 Conclusion

We presented TRACE, a TOD agent that retrieves relevant dialogue turn experiences conditioned on dialogue context and predicted belief states to guide both tool use and response generation. Experiments on MultiWOZ 2.1 and SGD demonstrate consistent improvements over competitive baselines, including strong generalization to unseen services. These results highlight the effectiveness of experience-guided retrieval for grounding agent behavior in TOD. Future work will explore tighter schema integration, more adaptive belief-state reasoning, and further improvements to retrieval quality.

Limitations

Our work targets TOD and is evaluated only on English benchmarks; performance may differ for other languages, domains, or open-ended conversational settings. Because TRACE relies on retrieving past dialogue snippets and on predicted belief states, errors or biases in the underlying dialogue corpus or state tracking can be propagated or amplified, leading to incorrect tool calls or inaccurate responses. If deployed with real user logs, retrieval from dialogue histories raises privacy concerns and should be accompanied by appropriate data governance (e.g., access controls, retention limits, and de-identification) and safeguards against the memorization of sensitive content.

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A Appendix

A.1 Dataset Statistics

Table 8 and Table 9 show the statistics of the MultiWOZ 2.1 and SGD datasets, respectively. The SGD dataset supports multiple services within a single domain, allowing for more diverse and fine-grained schema variations. In contrast, the MultiWOZ dataset is organized by domains, each associated with a single structured schema, resulting in more uniform slot configurations per domain.

A.2 Full Prompt for Belief State

Figure 5 shows the full prompt used for the belief state tracking stage. In this prompt, we provide the dialogue history, the cumulative belief state from prior turns, and the current user utterance. We add domain/service schemas that define the available

	#dialogs	#turns	avg. #turns	#slots	#domains	#services
Train	8,437	113,552	13.46	27	8	—
Dev	1,000	14,748	14.75	27	6	—
Test	1,000	14,744	14.74	26	5	—
	10,437	143,044	13.71	27	8	—

Table 8: The statistics of the MultiWOZ 2.1 dataset.

	#dialogs	#turns	avg. #turns	#slots	#domains	#services
Train	16,142	329,964	20.44	131	16	26
Dev	2,482	48,726	19.63	109	16	17
Test	4,201	84,594	20.14	119	18	21
	22,825	463,284	20.3	184	20	45

Table 9: The statistics of the SGD dataset.

slots and their descriptions. We include explicit output-format instructions with examples of valid and invalid belief states to enforce consistent formatting and accurate slot-value extraction.

A.3 Full Prompt for Dialogue Action

Figure 6 shows the full prompt used for the tool calls and response generation stage. We prompt the agent to decide whether to directly generate a system response or to invoke available tools. This decision is based on the current user utterance, the dialogue history, the list of available tools, and a set of reference dialogue turns. We clearly instruct the agent that the references are provided solely as guidance on how similar situations have been handled in the past, and that it should not copy or reuse any factual information from these references.

A.4 API Cost and Tokens

Tables 10 and 11 report API cost and token usage, broken down by user utterance, system response, retrieval query, and retrieval context, for AutoTOD and TRACE under simulation-based evaluation. Results suggest that TRACE uses more tokens due to its retrieval strategy, increasing per-dialogue API cost by approximately \$0.05–\$0.07 on MultiWOZ and \$0.08–\$0.11 on the SGD test dialogues. In addition, TRACE requires a longer runtime. However, the difference from AutoTOD is around 0.81–4.27 and 1.4–6.17 seconds on the MultiWOZ and SGD test set, respectively, per dialogue turn.

A.5 Actual Retrieved Dialogue Snippets and Generated Responses

A.5.1 Dialogue Retrieval Examples

Tables 12 and 13 present the actual test dialogue from the MultiWOZ and SGD datasets. Each di-

Belief State Prompt
You are a belief state tracker for task-oriented dialogue systems. Your job is to extract the updated user belief state at the current turn based on: - The conversation history - The cumulative belief state based on the history - The new user utterance A belief state consists of slot-value pairs that capture the user's stated constraints or preferences, grouped by domain (e.g., hotel, train, restaurant). # Instructions: 1. Extract the current belief state. 2. The cumulative belief state is provided as a guidance. It can be wrong. Please verify it. 3. The provided cumulative belief state is a reference and may contain errors. Please verify it. 4. If the user has not yet provided any constraint, return <code>-</code>. For MultiWOZ -> [5. This key-value pair, 'area: Cambridge', should be completely ignored] For SGD -> [5. Today is 2019-03-01, Friday. (This Saturday is 2019-03-02. This Sunday is 2019-03-03.)] # Output format: domain1 (slot1: value1, slot2: value2), domain2 (slot1: value1) ## Valid example output: hotel (price: cheap, internet: yes, people: 4), restaurant (food: italian) ## Invalid example output: hotel (price: cheap), hotel (internet: yes, people: -), restaurant (-) # Valid domains and slots: {service_schemas} Start belief state tracking! Conversation history: {chat_history} Belief state of the history: {belief_history} User: {user_input} New belief state: {llm_output}

Figure 5: The complete prompt for the belief state tracking in TRACE.

logue turn is presented with the user utterance, predicted belief state, top-3 dialogue turns, and system response.

A.5.2 Comparison of Actual Responses based on Simulation-based Evaluation

Table 14 shows the actual responses generated by the baseline agent, AutoTOD, and our proposed agent, TRACE, for a MultiWOZ test dialogue. Both using GPT-4o-mini as the underlying LLM. Based on these results, TRACE successfully provided accurate information across both domains and completed the train booking according to the user's preferences. These results support our hypothesis that TRACE benefits from the retrieved dialogue snippets, which demonstrate how tool calls should be constructed based on the belief states. The retrieved examples also help the agent recognize what additional information needs to be requested to complete the task. For instance, in the first dialogue turn, TRACE correctly asked

Method	Total cost (\$)	Avg. per dialogue			User utt.	Response	Avg. per turn		Run time (s)*
		Cost (\$)	# Turns	Run time (s)*			Ret. query	Ret. context	
AutoTOD									
Llama-3.1-70B	–	–	13.24	276.15	44.54	64.52	–	–	20.67
Qwen2.5-72B	–	–	5.26	76.54	32.15	103.23	–	–	15.1
GPT-3.5	19.32	0.019	5.41	18.37	20.92	60.99	–	–	3.23
GPT-4o-mini	14.85	0.015	11.68	82.27	26.62	85.07	–	–	8.52
TRACE									
Llama-3.1-70B	–	–	12.95	319.96	45.14	60.23	82.46	228.58	24.94 (+4.27)
Qwen2.5-72B	–	–	5.4	85.28	30.71	97.41	57.98	228.73	16.21 (+1.1)
GPT-3.5	24.44	0.024	5.29	21.14	21.13	69.3	47.53	220.60	4.04 (+0.81)
GPT-4o-mini	21.94	0.022	11.18	82.02	26.58	84.96	59.64	238.91	10.46 (+1.94)

Table 10: Comparison of the API cost, number of dialogue turns, tokens of user utterance, response, retrieval query, and retrieval context (k=3) by AutoTOD and TRACE on 1,000 MultiWOZ test dialogues under the simulation-based evaluation setting. * denotes the values obtained by ablation studies of 100 dialogues.

Method	Total cost (\$)	Avg. per dialogue			User utt.	Response	Avg. per turn		Run time (s)*
		Cost (\$)	# Turns	Run time (s)*			Ret. query	Ret. context	
AutoTOD									
Llama-3.1-70B	–	–	11.47	17.13	37.89	42.59	–	–	7.14
Qwen2.5-72B	–	–	7.26	35.23	32.28	81.38	–	–	11.78
GPT-3.5	6.73	0.007	3.96	7.87	17.83	62.03	–	–	2.19
GPT-4o-mini	6.59	0.007	10.9	44.75	23.86	69.59	–	–	4.2
TRACE									
Llama-3.1-70B	–	–	12.53	79.42	34.97	52.82	97.70	263.71	11.12 (+3.98)
Qwen2.5-72B	–	–	7.45	138.4	30.61	77.44	79.96	263.15	17.95 (+6.17)
GPT-3.5	17.52	0.018	5.22	15.09	18.4	56.65	50.93	220.72	3.59 (+1.4)
GPT-4o-mini	15.35	0.015	10.16	66.74	24.28	77.95	83.45	264.9	6.69 (+2.49)

Table 11: Comparison of the API cost, number of dialogue turns, tokens of user utterance, response, retrieval query, and retrieval context (k=3) by AutoTOD and TRACE on 1,000 SGD test dialogues under the simulation-based evaluation setting. * denotes the values obtained by ablation studies of 100 dialogues.

the user to specify the departure location and time for the train journey, while AutoTOD prematurely returned train options with incorrect destinations.

On SGD, the example responses by AutoTOD and TRACE are shown in Table 15. The results show similar trends in which the baseline agent could not provide any options based on the user’s preferences for the RentalCars domain. Meanwhile, TRACE successfully found some options and offered them to the user correctly.

A.5.3 Comparison of Actual Responses based on Static-User Evaluation

Table 16 presents the responses generated by both AutoTOD and our proposed agent, TRACE, using GPT-4o-mini, for a MultiWOZ test dialogue. In this example, TRACE provided verifiable information that leads to a successful restaurant reservation. In contrast, AutoTOD appeared to hallucinate entity details, which results in confusion and ultimately a failure to complete the booking within the fixed set of user turns.

Table 17 shows the responses generated by AutoTOD and our proposed agent, TRACE, using GPT-

4o-mini on a test dialogue from the SGD dataset. In this example, TRACE successfully identified a restaurant that meets the user’s criteria and completes the reservation. In contrast, AutoTOD failed to retrieve any suitable restaurant and appears to hallucinate by responding as if it had already offered options. This is reflected in the user’s fix utterance, “I’m glad to hear that you love the place”, despite no restaurant options having been presented earlier in the dialogue.

A.6 Human Evaluation

We conduct human evaluation on a subset of test dialogues from both MultiWOZ and SGD. This evaluation is designed as a dialogue-level task-correctness assessment rather than a subjective preference judgment: annotators read the full dialogue interaction, compare the predicted outcome with the user goal of each domain, and verify whether the requested constraints are satisfied using the corresponding database records. Due to the high annotation cost of this verification process, each dialogue is rated by one trained annotator. To mitigate the risk of inconsistent judgments, we used

Dialogue Action Prompt

The AI Assistant is an intelligent agent that helps the user complete complex tasks, which may involve several sub-tasks.

The AI Assistant will receive the user's utterance with a set of relevant past conversation turns provided as references. The AI Assistant MUST first identify the sub-tasks implied by the user's request and determine the necessary next actions required to fulfill the request. The AI Assistant MUST use the appropriate tools to gather accurate and up-to-date information needed to proceed.

Important Guidelines:

- The reference conversation turns are provided only to give contextual insight into how similar requests were handled in the past.
- Do not reuse or rely on specific content from the references in the actual response.
- Any factual or task-related information must be verified using tools, not inferred from the references.

The AI Assistant is always friendly and polite.

Task Overall:
For MultiWOZ -> [#Domain, #Description, #Tools, #Logic]
For SGD -> [#Service, #Description, #Function and info]

Output Format Instructions

To use an Tool, please use the following format:

```

...
Thought: Do I need to use a tool? Yes
Action: the tool name to use
Action Input: the input to the tool
Observation: [leave empty for the tool output]
...

```

- Available tool names: Restaurant Query, Restaurant Reservation, Hotel Query, Hotel Reservation, Attraction Query, Train Query, Train Tickets Purchase, Taxi Reservation
- The content of Action Input can only come from the user's utterance, and there cannot be any imaginary content irrelevant to the user's utterance.
- If the user asks for some information of a venue (restaurant, hotel, attraction) or train without specifying the name or trian id, the AI Assistant should ask for the name/trian id from the user first.

When you have a response to say to the User, or if you do not need to use an Tool, you MUST use the format:

```

...
Thought: Do I need to use a tool? No
AI Assistant: [your response here]
...

```

- The final response of the AI Assistant should summary all the previous Action steps and Observations. The response should be independent and not rely on previous contents.

Begin!

Previous conversation history:
{chat_history}

New input: {user_input}

{llm_output}

Figure 6: The complete prompt for the tool calls and response generation.

an expert-calibrated protocol: the annotator was trained on pilot examples with an expert familiar with TOD evaluation, and the annotation guidelines were refined until consistent judgments were reached. The full evaluation was then conducted using this calibrated protocol. The human evaluation interface, including the instructions and evaluation tasks, is shown in Figure 7.



Figure 7: Human evaluation interface.

User: Hi, I am trying to plan a trip and could use some help with the trains.

Belief State: restaurant (name: Restaurant Two Two)

Reference 1:

User: i am looking for a particular restaurant. Its name is called restaurant two two.

Belief state: restaurant (name: restaurant two two)

Assistant: Restaurant Two Two is an expensive french restaurant on the north side. Would you like to make a reservation?

Reference 2:

User: I'm looking for a restaurant called restaurant two two?

Belief state: restaurant (name: restaurant two two)

Assistant: Restaurant Two Two is located at 22 Chesterton Road Chesterton. Would you like to book a reservation?

Reference 3:

User: Would you be able to help me locate a restaurant in Cambridge?

Belief state: -

Assistant: We have many great places to eat in Cambridge. What kind of cuisine or price range are you looking for?

TRACE: Restaurant Two Two is located at 22 Chesterton Road, Chesterton, Cambridge, with a postcode of CB4 3AX. It's in the north area of Cambridge and offers French cuisine. The restaurant is categorized as expensive. If you need more information or would like to make a reservation, let me know!

User: That would be great. I need to book a table for 8 people on Friday. If possible, I'd prefer a table at 11:15, but 10:15 is also acceptable.

Belief State: restaurant (name: Restaurant Two Two, people: 8, day: Friday, time: 11:15)

Reference 1:

User: That would be great. Can you book a table for 3 at 11:45 on Friday?

Belief state: restaurant (name: restaurant one seven, time: 11:45, day: friday, people: 3)

Assistant: All right, I've got you booked for Friday at 11:45 for 3 people. Would you like the reference number?

Reference 2:

User: Can you book a table for 8 people at 11:00 on thursday?

Belief state: restaurant (name: restaurant two two, people: 8, day: thursday, time: 11:00)

Assistant: Yes, your ref. # is E7D8F4SM. Can I help you with anything more?

Reference 3:

User: That sounds great. Can you book a table for 8 people on Friday at 13:30?

Belief state: restaurant (food: british, pricerange: moderate, name: restaurant one seven, area: centre, people: 8, day: friday, time: 13:30),

Assistant: I have made those reservations. Your reference number is F9G08K4X.

TRACE: I have successfully booked a table for 8 people at Restaurant Two Two for this Friday at 11:15. Your reservation reference number is blr90xek. If you need any further assistance, feel free to ask!

User: Thanks. Can you help me find a train, too? I want to leave Cambridge after 12:15 on Saturday and head to Peterborough.

Belief State: train (departure: Cambridge, destination: Peterborough, leave: after 12:15, day: Saturday)

Reference 1:

User: I'm looking for a train for my trip to Cambridge. I need it to leave on Monday and to depart from Peterborough. Can you give me some times?

Belief state: train (destination: cambridge, day: monday, departure: peterborough)

Assistant: Sure, there are plenty of available time starting at 05:19 and going all day. Was there a particular time of day you wanted to depart?

Reference 2:

User: I'm looking for a train for my trip to Cambridge. I need it to leave on Monday and to depart from Peterborough. Can you give me some times?

Belief state: train (destination: cambridge, day: monday, departure: peterborough)

Assistant: Sure, there are plenty of available time starting at 05:19 and going all day. Was there a particular time of day you wanted to depart?

Reference 3: User: I would be in cambridge and I need to find a train that will be leaving from peterborough this sunday, can you help me find one?

Belief state: train (destination: cambridge, day: sunday, departure: peterborough),

Assistant: Am I correct that you are travelling from Peterborough to Cambridge on Sunday? There are plenty of options for that trip. Do you have any preferences on departure and arrival times?

TRACE: For your travel from Cambridge to Peterborough on Saturday, leaving after 12:15, here are some available trains:

- Train ID TR3934, leaving at 12:34 and arriving at 13:24.
- Train ID TR6164, leaving at 13:06 and arriving at 13:56.
- Train ID TR9212, leaving at 13:34 and arriving at 14:24.
- Train ID TR2834, leaving at 14:06 and arriving at 14:56.
- Train ID TR1145, leaving at 14:34 and arriving at 15:24.

Would you like to book a ticket for any of these trains? If so, please specify the train ID and the number of tickets you need.

Table 12: The first three dialogue turns between the user simulator and TRACE for the test MultiWOZ dialogue, MUL1575, with Qwen2.5. For each user utterance, TRACE tracks the belief state and retrieves the relevant past dialogue turns from the corpus as references for generating tool calls and responses. Note that we only present the system response at each turn, without the actual tool calls and tool results.

User: I am looking for some attractions to visit. Belief State: - Reference 1: User: I am looking for some attractions to visit Belief state: - Function calls: - Assistant: Where are you looking to visit? Reference 2: User: I am looking for some attractions to visit. Belief state: - Function calls: - Assistant: Which city should I search in? Reference 3: User: I'm looking for some attractions to visit. Belief state: - Function calls: - Assistant: What city do you want to search in? TRACE: What city do you want to search in?
User: I prefer attractions in London. Belief State: travel (location: london) Reference 1: User: I want attractions in London, England. Belief state: Travel (location: London, England) Function calls: FindAttractions (location: London) Assistant: You should check out 30 St Mary Axe (The Gherkin). This is a Historical Landmark in the city of London, England. Reference 2: User: I am interested about attractions in London, UK. Belief state: Travel (location: London, UK) Function calls: FindAttractions (location: London) Assistant: There are 10 attractions which match what you asked for. You might consider visiting 30 St Mary Axe (The Gherkin), which is a Historical Landmark. Reference 3: User: I am looking for attractions in London, UK. Belief state: Travel (location: London, UK) Function calls: FindAttractions (location: London) Assistant: I found 10 attractions you may like. How about 30 St Mary Axe (The Gherkin)? It is a Historical Landmark. TRACE: I found several attractions in London that you might enjoy. Here are a few options: 1. **30 St Mary Axe (The Gherkin)** - A Historical Landmark. Free entry and good for kids. 2. **Alexandra Palace** - A Performing Arts Venue. Paid entry but good for kids. 3. **Alexandra Palace Ice Rink** - A Sports Venue. Free entry and good for kids. 4. **All England Lawn Tennis & Croquet Club** - A Sports Venue. Paid entry but good for kids. Would you like more details on any of these, or should I provide more options?
User: Can you confirm if the entrance to 30 St Mary Axe (The Gherkin) is actually free and if it's a good place to take kids? Belief State: travel (location: london, attraction_name: 30 St Mary Axe (The Gherkin), free_entry: true, good_for_kids: true) Reference 1: User: Can you tell me if this attraction is free or if it has an entrance fee? Belief state: Travel (good_for_kids: True, location: London) Function calls: - Assistant: There is no fee to get into this attraction. Reference 2: User: That'll work. Can I book a cab to get there please? Belief state: Travel (good_for_kids: True, location: London, England, free_entry: True), Weather (city: London, England), RideSharing (destination: 30 St Mary Axe (The Gherkin)) Function calls: - Assistant: Absolutely. Is a shared ride okay for you? Reference 3: User: No, not right now. Get me a cab to go there. Belief state: Travel (free_entry: True, good_for_kids: True, location: London, UK), Calendar (event_date: 1st of March), RideSharing (destination: 30 St Mary Axe (The Gherkin)) Function calls: - Assistant: Do you prefer a shared ride and how many people are going? TRACE: Based on the information I have, the entrance to 30 St Mary Axe (The Gherkin) is indeed free, and it is considered a good place to take kids. Would you like more details about this attraction or any other recommendations?

Table 13: The first three dialogue turns between the user simulator and TRACE for the test SGD dialogue, test_9_00022, with Qwen2.5. For each user utterance, TRACE tracks the belief state and retrieves the relevant past dialogue turns from the corpus as references for generating tool calls and responses. Note that, we only present the system response at each turn, without the actual tool calls and tool results.

Goals: You are planning your trip in Cambridge. You are looking for a **train**. The train should depart from **cambridge** and should go to **peterborough**. The train should **leave after 18:45** and should leave on **wednesday**. Once you find the train you want to make a booking for **4 people**. Make sure you get the **reference number**. You are also looking for **places to go** in town. The attraction should be in the type of **museum** and should be in the **centre**. Make sure you get **entrance fee** and **address**

User: Hi, I am trying to plan a trip and could use some help with the trains.

AutoTOD: You can take the following trains from Cambridge to London Kings Cross on Monday after 08:05: TR7409 at 09:00, TR1111 at 11:00, TR6110 at 13:00, TR6028 at 15:00, and TR7786 at 17:00. If you need further assistance with purchasing tickets or have any other questions, feel free to ask!

User: I'm actually looking for a train from Cambridge to Peterborough on Wednesday after 18:45, not to London. Can you help me find that?

AutoTOD: Here are some recommended trains from Cambridge to Peterborough leaving after 18:45 on Wednesday: TR8199 at 19:06, TR0797 at 19:34, TR5604 at 20:06, TR5117 at 20:34, and TR4188 at 21:06. All trains have a duration of 50 minutes and a price of 16.50 pounds. If you would like to purchase tickets or need any further assistance, feel free to let me know!

User: I'd like to book the TR8199 train for 4 people, please.

AutoTOD: I have successfully booked 4 tickets for the TR8199 train from Cambridge to Peterborough. Your reference number for the booking is cegz25wu. If you have any other questions or need further assistance, feel free to ask!

User: I'm also interested in finding a museum in the center of Cambridge. Can you recommend one and provide the entrance fee and address?

AutoTOD: You can explore several museums in the centre of Cambridge. Noteworthy options include:

1. **Broughton House Gallery** - Address: 98 King Street, Free Entry
2. **Museum of Archaeology and Anthropology** - Address: University of Cambridge, Downing Street. Free Entry, Open Tuesday to Saturday from 10:30 a.m. to 4:30 p.m.
3. **Fitzwilliam Museum** - Address: Trumpington Street. Free Entry with varied hours.
4. **Whipple Museum of the History of Science** - Address: Free School Lane. Free Entry, Open Monday to Friday from 12:30 p.m. to 4:30 p.m.

If you would like more information or help with anything else, just let me know!

User: I'd like to gather some information on one of those museums. Can you provide the entrance fee and address for the Museum of Archaeology and Anthropology?

AutoTOD: The Museum of Archaeology and Anthropology is located at the University of Cambridge, Downing Street. The entrance is free, and the museum is open Tuesday to Saturday from 10:30 a.m. to 4:30 p.m. If you have any more questions or need further assistance, feel free to ask!

- Dialogue Ends -

User: Hi, I am trying to plan a trip and could use some help with the trains.

TRACE: I'd be happy to help you with planning your train trip! To start, could you please provide me with your departure location, destination, and any time preferences you may have for your journey?

User: I'm leaving Cambridge and heading to Peterborough on Wednesday after 18:45.

TRACE: The available trains departing from Cambridge to Peterborough on Wednesday after 18:45 include:

- TR8199 at 19:06, arriving at 19:56
- TR0797 at 19:34, arriving at 20:24
- TR5604 at 20:06, arriving at 20:56
- TR5117 at 20:34, arriving at 21:24
- TR4188 at 21:06, arriving at 21:56

If you would like to purchase tickets for any of these trains, please let me know the train ID and the number of tickets!

User: I would like to book the TR8199 at 19:06 for 4 people, please.

TRACE: Your booking for the train TR8199 at 19:06 from Cambridge to Peterborough for 4 people has been successful! Your reference number is zfvv3a2h. If you need any more assistance or have further questions, feel free to ask!

User: I'm also looking for a museum in the center of town. Can you help me with that?

TRACE: I recommend visiting the Broughton House Gallery, which is located at 98 King Street. If you need any further assistance or have more questions, feel free to ask!

User: Could you please provide the entrance fee for the Broughton House Gallery?

TRACE: The Broughton House Gallery is a museum located at 98 King Street in the center of town, and the entrance fee is free! If you have any other questions or need further assistance, please let me know!

- Dialogue Ends -

Table 14: The actual responses for the test MultiWOZ dialogue, PMUL1463, by AutoTOD and TRACE with GPT-4o-mini as the LLM using the *simulated user*. Although AutoTOD provided the information of the attraction (*i.e.*, Museum of Archaeology and Anthropology) to the user, the information cannot be verified from the actual database. Thus, AutoTOD received zero scores for both Inform and Success in the attraction domain. In contrast, TRACE correctly provided information for both domains and successfully booked the train according to the user's preferences.

Goals: Get and reserve the available cars at Seattle for the start date of the 9th of March, pickup time at eleven in the morning, end date of the 10th of this month. The car type should be Sedan with a pickup location of King Street Station and without insurance. The rental cost needs to be provided.

User: Can you help me rent a car?

AutoTOD: Sure! I can help you with that. Could you please provide me with the following details?

1. The city where you want to rent the car.
2. The start date for the rental.
3. The pickup time.
4. The end date for the rental.
5. The type of car you prefer (Hatchback, Sedan, or SUV).

Once I have this information, I can find available cars for you!

User: I want to rent a car in Seattle, WA. I will need it from the 9th of March to the 10th of this month. I would like to pick it up at eleven in the morning.

AutoTOD: I can help you with that! Just to confirm, you want to rent a car in Seattle, WA from March 9th to March 10th, with a pick-up time of 11:00 AM. Could you please specify the type of car you would like? Your options are: Hatchback, Sedan, SUV. Also, would you like to purchase insurance for the rental?

User: I don't have a specific type of car in mind, so any available option would be fine. And no, I don't need insurance for the rental.

AutoTOD: It seems that there are no cars available for rent in Seattle, WA from March 9th to March 10th at 11:00 AM. Would you like to try a different location or adjust the dates?

User: I would like to stick with Seattle, WA and the same dates. Can you please check again for available cars?

AutoTOD: It seems that there are no available cars for rent in Seattle, WA from March 9th to March 10th at 11 AM. Would you like to try a different location or adjust the dates?

- The Dialogue Turn Context Repeated until Maximum Number of Turn Reaches -

User: Can you help me rent a car?

TRACE: Sure! I can help you with that. To proceed, I need a bit more information from you:

1. What city do you want to rent the car in?
2. What are the start and end dates for the rental?
3. What time do you want to pick up the car?
4. What type of car are you looking for? (Options: Hatchback, Sedan, SUV)
5. Would you like to add insurance to your rental?

Once I have this information, I can find available cars for you!

User: I want to rent a car in Seattle, WA. I will need it from the 9th of March to the 10th of this month. I would like to pick it up at Eleven in the morning.

TRACE: Thank you for the details! Just to confirm before I search for available cars: **City:** Seattle, WA, **Pickup Date:** March 9, 2019, **Pickup Time:** 11:00 AM, **End Date:** March 10, 2019, **Car Type:** (Please specify if you have a preference for Hatchback, Sedan, or SUV), **Add Insurance:** (Would you like to add insurance to your rental?)

Once I have the car type and insurance preference, I can find available cars for you!

User: I don't have a specific car type in mind, so you can just find any available options. Also, I don't need insurance for the rental.

TRACE: I found several options for you to rent a car in Seattle from March 9 to March 10. Here are the available cars:

1. **Car Type:** Sedan, **Car Name:** Audi A8, **Pickup Location:** King Street Station, **Price per Day:** \$32.00
2. **Car Type:** SUV, **Car Name:** Citroen C5, **Pickup Location:** Seattle-Tacoma International Airport, **Price per Day:** \$51.00
3. **Car Type:** SUV, **Car Name:** Dacia Duster, **Pickup Location:** King Street Station, **Price per Day:** \$40.00

Please let me know which car you would like to reserve, and I will proceed with the booking!

User: I would like to reserve the Audi A8 from King Street Station, please.

TRACE: To confirm your reservation for the Audi A8, here are the details: **Car Type:** Sedan, **Car Name:** Audi A8, **Pickup Location:** King Street Station, **Pickup Date:** March 9, 2019, **Pickup Time:** 11:00 AM, **End Date:** March 10, 2019, **Add Insurance:** No

Shall I proceed with the reservation?

User: Yes, please proceed with the reservation. Also, could you let me know the cost for renting the car per day?

TRACE: Your reservation for the Audi A8 has been successfully completed! Here are the details: **Car Type:** Sedan, **Car Name:** Audi A8, **Pickup Location:** King Street Station, **Pickup Date:** March 9, 2019, at 11:00 AM, **End Date:** March 10, 2019, **Price per Day:** \$32.00, **Reference Number:** lz5g0xtv

If you have any more questions or need further assistance, feel free to ask. Enjoy your trip!

User: Thank you for the reservation details! Just to confirm, the cost for renting the Audi A8 is \$32.00 per day, correct?

TRACE: Yes, that's correct! The cost for renting the Audi A8 is \$32.00 per day. If you have any more questions or need further assistance, feel free to ask. Enjoy your rental!

User: Thank you for your help! That covers everything I needed.

TRACE: You're very welcome! I'm glad I could assist you. If you need help in the future, don't hesitate to reach out. Have a great day and enjoy your rental! Goodbye!

- Dialogue Ends -

Table 15: The actual responses for the test SGD dialogue, test_20_00076, by AutoTOD and TRACE with GPT-4o-mini as the LLM using the *simulated user*. While AutoTOD failed to provide any suitable entities to the users, even though the user repeatedly stated their preferences, TRACE successfully provided the car details and reserved the car rental according to the user's preferences.

Goals: You are looking for a **place to dine**. The restaurant should be in the **moderate** price range and should serve **portuguese** food. If there is no such restaurant, how about one that serves **turkish** food. Once you find the **restaurant** you want to book a table for **1 people** at **14:00** on **monday**. Make sure you get the **reference number**

User: Are there any Portuguese restaurants in Cambridge?

AutoTOD: The recommended Portuguese restaurants in Cambridge are "Nandos" located at "33-34 Saint Andrews Street" with a price range of "cheap", and another "Nandos" at "Cambridge Leisure Park Clifton Way," also with a "cheap" price range. Would you like to know more about them or make a reservation?

User: If one of them has a moderate price range please give me that address. If not tell me about Turkish restaurants instead.

AutoTOD: There are several Turkish restaurants in the centre of Cambridge. Notable options include 'Efes' located at "King Street" and 'Anatolia' at "30 Bridge Street," both offering moderate dining experiences. Additionally, there is 'Meze Bar' at "196 Mill Road," which is more expensive. Would you like more information about any of these restaurants or to make a reservation?

User: Actually I need a moderately priced restaurant. Are there any fitting that description?

AutoTOD: I recommend the following moderate-priced restaurants in Cambridge: ****De Luca Cucina and Bar****, located at 83 Regent Street, serving modern European cuisine with a signature roasted Barbary duck breast; ****Galleria****, at 33 Bridge Street, offering European dishes, including poached fillets of monkfish; and ****Cow Pizza Kitchen and Bar****, situated on Corn Exchange Street, known for its unique gastropub experience with a wasabi spiked hamburger. Would you like more information or to make a reservation at any of these restaurants?

User: Yes, I need a reservation for 1 people at 14:00 on monday.

AutoTOD: It seems I cannot find "De Luca Cucina and Bar" in my database. Would you like to make a reservation at one of the other recommended moderate-priced restaurants, such as ****Galleria**** or ****Cow Pizza Kitchen and Bar****? Please let me know!

User: Thanks so much. That's all for today. Bye!

AutoTOD: You're very welcome! If you have any more questions in the future or need assistance, feel free to reach out. Have a wonderful day and goodbye!

- Dialogue Ends -

User: Are there any Portuguese restaurants in Cambridge?

TRACE: The only Portuguese restaurant in the centre of Cambridge is Nandos City Centre, located at 33-34 Saint Andrews Street. It offers Portuguese cuisine at a cheap price. If you would like to make a reservation or need more assistance, just let me know!

User: If one of them has a moderate price range please give me that address. If not tell me about Turkish restaurants instead.

TRACE: I found two Turkish restaurants with moderate pricing: Efes located on King Street City Centre and Anatolia at 30 Bridge Street City Centre. If you would like to make a reservation or need further assistance, just let me know!

User: Actually I need a moderately priced restaurant. Are there any fitting that description?

TRACE: I found two moderately priced Turkish restaurants: Efes located at King Street City Centre and Anatolia at 30 Bridge Street City Centre. If you would like to make a reservation or need further assistance, just let me know!

User: Yes, I need a reservation for 1 people at 14:00 on monday.

TRACE: Your reservation for 1 person at Anatolia for Monday at 14:00 has been successfully booked. The reference number for your reservation is jolln2hg. If you need any further assistance, feel free to ask!

User: Thanks so much. That's all for today. Bye!

TRACE: You're welcome! If you have any more questions in the future, don't hesitate to ask. Have a wonderful day and enjoy your time at the restaurant! Goodbye!

- Dialogue Ends -

Table 16: The actual responses for the test MultiWOZ dialogue, SNG01608, by AutoTOD and TRACE with GPT-4o-mini as the LLM using the *fixed user utterances* at every turn. Based on these responses, AutoTOD could not successfully book the restaurant for the user. Meanwhile, TRACE successfully booked the restaurant according to the user's preferences.

Goals: Search and book house for 1 adult in new york city. The phone number and reference number needs to be provided.

User: you have to help me in finding a house for 1 person

AutoTOD: Sure, I can help you with that. Could you please provide me with the location where you would like to find a house?

User: new york city

AutoTOD: It seems that there are no available houses in New York City for 1 person at the moment. Would you like me to search for houses in a different location or with different criteria?

User: send me their phone number

AutoTOD: I'm sorry, but I couldn't find any houses in New York City. Would you like me to search for houses in a different location?

User: I love the place already

AutoTOD: I'm glad to hear that you love the place! Would you like to proceed with booking the house in New York City for 1 person?

User: No, I don't need you to.

AutoTOD: It seems like there are no available houses in New York City for 1 person at the moment. Would you like me to search for houses in a different location or with different criteria?

User: no, it's okay

AutoTOD: I'm glad you love the place! If you need any further assistance or have any other requests, feel free to let me know. Have a great day! Goodbye!

- Dialogue Ends -

User: you have to help me in finding a house for 1 person

TRACE: I have found several houses in New York that are suitable for 1 person. Would you like more information about a specific house, or should I proceed with booking one for you?

User: new york city

TRACE: I'm sorry, but I couldn't find any houses in New York City that match your criteria for 1 person. Would you like to try a different location or adjust the search criteria?

User: send me their phone number

TRACE: I have found some houses in New York City for 1 person. Here are the phone numbers for a few of them:

1. Address: 102 West 57th Street, Phone Number: +1 212-379-0103
2. Address: 100 Greenwich Street, Phone Number: +1 917-409-0800
3. Address: 100-15 Ditmars Boulevard, East Elmhurst, NY 11369, USA, Phone Number: +1 718-512-0248

Please let me know if you need more information or if you would like to proceed with booking one of these houses.

User: I love the place already

TRACE: Great choice! I have successfully booked the house for you in New York City for 1 person. Your reference number is namexzbp. Is there anything else I can assist you with?

User: No, I don't need you to.

TRACE: Thank you for using the service! If you need any help in the future, feel free to ask. Have a great day! Goodbye!

User: no, it's okay

TRACE: Thank you for using the service! If you need any assistance in the future, feel free to ask. Have a wonderful day! Goodbye!

- Dialogue Ends -

Table 17: The actual responses for the test SGD dialogue, test_10_00111, by AutoTOD and TRACE with GPT-4o-mini as the LLM using the *fixed user utterances* at every turn. In this example, AutoTOD failed to identify a suitable house for the user, while TRACE successfully finds and books the correct one. TRACE could also provide the phone number and reference number. Additionally, AutoTOD exhibits mild hallucination as it mistakenly assumes it has already presented housing options, when the fixed user response is that they love the house, despite not being shown any options.